

Assignment 1

Ziyan Liu 2025/10/29

Problem 1

For random a, b and c, the results are as below,

$$[x, y, z] = [9, 7, 2] \rightarrow x + y - 10z = -4$$

$$[x, y, z] = [3, 1, 5] \rightarrow x + y - 10z = -46$$

And for a = 5, b = 15, c = 10, the result is as below.

$$[x, y, z] = [10, 5, 15] \rightarrow x + y - 10z = -135$$

Problem 2

According to the continuous ceiling function defined as $F(x) = F(\text{ceil}(x/3)) + 2x$, where $F(1) = 1$, the list with both x and $F(x)$ is as below.

x	F(x)
1	1
2	5
3	7
10	33
28	89

Problem 3

3.2 As to the the number of ways for any x from 10 to 60, assign the number of ways to a list called Number_of_ways, so which x yields the maximum of Number_of_ways?

x_values : [10, 11, 12, 13, 14] ... [56, 57, 58, 59, 60]

Number_of_ways : [1, 10, 55, 220, 715] ... [715, 220, 55, 10, 1]

max ways : 4395456

x with max ways : [35]

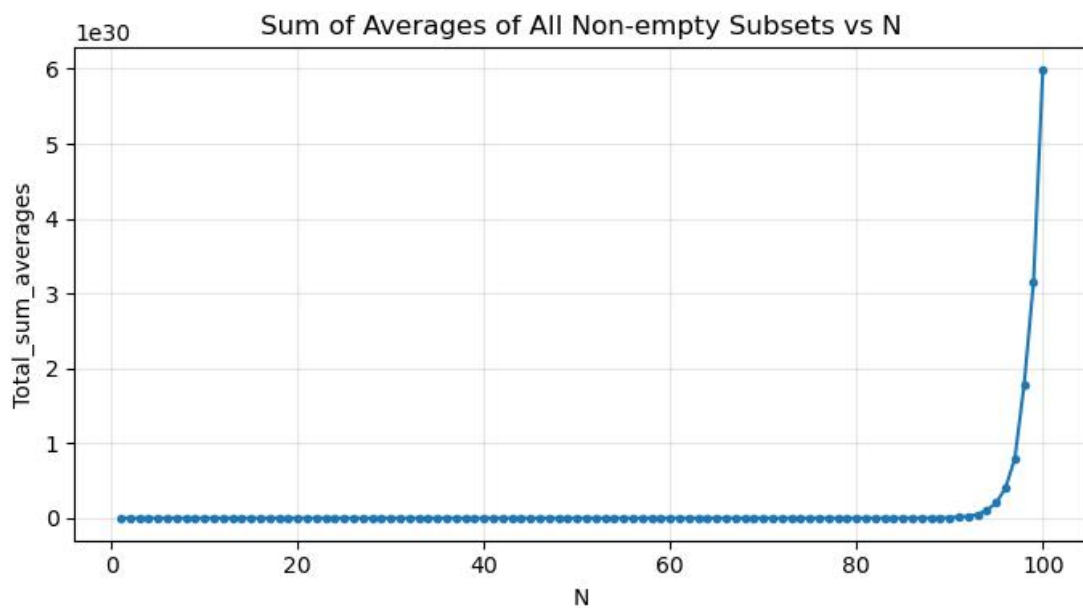
Problem 4

4.3 Observation:

- The overall trend increases with N, the curve rises exponentially, maybe because the formula is :

$$\text{Sum(arr)} = (2^N - 1)/N$$

the growth is mainly driven by 2^N .



Problem 5

5.1 Sample grid (1 ok / 0 blocked):

```
[[1 1 1 1 0 0]
 [0 0 1 0 0 1]
 [0 1 0 1 0 1]
 [1 1 1 1 0 0]
 [0 1 0 0 0 1]]
```

5.2 Paths on sample grid: 0

5.3 Mean number of paths over 1000 runs (N=10, M=8): 0.466

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